



NAKUJA PROJECT

Solid Propulsion Team

KNSB Grain Curing & Testing — Data Sheet

Batch Identification

Batch ID	
Date	
Propellant type	
Test condition	☐ A — Baseline ☐ B — Surfactant only ☐ C — Vacuum only ☐ D — Surfactant only
Personnel	
Notes / remarks	

Ingredient Masses

Ingredient	Target mass (g)	Actual mass (g)
Sorbitol	690.87	
Potassium nitrate (fine)	944.84	
Surfactant (SLS or equiv.)	0.87	
Total dry mix	1635.71	

Surfactant dose used	g per 100 g dry mix (limit: 0.053 g/100 g)
KNO ₃ condition	☐ Fine (as-received) ☐ Ground (low setting)



Mixing & Casting Log

Step	Target	Actual / observed
Sorbitol heat-up	150°C, 30 min	
Caramelisation temp.	Reduce to 120°C	
Caramel colour change	Confirmed at	
Surfactant addition	At 90°C, stir 1 min	
KNO ₃ addition start	Reduce to 90°C	
Return to temp.	120°C after 1 min	
Post-mix duration	20 min at 120°C	
Cool-down	Reduce to 60°C	
Casting completed at	—	

Mix observations	
Casting observations	

Vacuum Degassing

Degassing performed?	☐ Yes ☐ No
Time since casting	min (target: within 30–60 min)
Pump used	VE115N ☐ Confirmed on nameplate

Parameter	Target	Actual
Working vacuum	10–50 mbar	mbar
Pump-down time	≈100 s (idealised)	s
Dwell time	10–15 min (or until bubbling stops)	min
Vent-back method	Gradual (slow crack of vent valve)	☐ Gradual ☐ Fast

Bubbling behaviour	
Anomalies (foaming, boil-over, leaks)	



Cure

Cure start date/time	
Cure end date/time	
Total cure duration	days (target: 5–7 days)
Storage conditions	

Post-Cure Quality Checks

Density and Porosity Estimate

Measurement	Expected / reference	Measured
Grain mass	—	g
Grain volume (geometric)	—	cm ³
Measured density, ρ_{actual}	—	g/cm ³
Theoretical density, ρ_{theor}	1.81 g/cm ³	1.81 g/cm ³
Density ratio $\rho_{\text{actual}}/\rho_{\text{theor}}$	Target: as high as possible	%

Visual Inspection (Cross-Section)

Cross-section taken?	☐ Yes ☐ No
Void / microcrack presence	☐ None ☐ Minor ☐ Significant
Surface finish (bore)	☐ Smooth ☐ Rough ☐ Irregular
Colour uniformity	☐ Uniform ☐ Banding ☐ Inclusions
General observation	



Overall Assessment

Grain accepted?	☐ Yes — proceed to static fire ☐ No — discard / investigate
Anomalies noted	
Recommended follow-up	
Sign-off	Prepared by: _____ Reviewed by: _____ Approved by: _____ Date: _____

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